

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)		The most stable nuclei (or reference to elements near to peak i.e. Fe, Ni Ca) are therefore where the curve (or binding energy per nucleon) reaches its maximum. (1) Nuclei of small atomic mass number (lhs of graph) can combine to produce species of larger atomic mass number, [hence larger binding energy per nucleon. Energy is released]. Fusion. (1) Nuclei of large atomic mass number (on rhs of graph) break down to produce species of smaller atomic mass number, [hence larger binding energy per nucleon. Energy is released]. Fission. (1) Reference anywhere to there being energy released, when a <u>reduction in mass</u> occurs i.e. mass converted to energy. (1)	4			4		
	(b)		Mass defect = $4(1.00728) + 2(0.00055)$ (1) $- 4.00151 = [0.02871 \text{ u}]$ (1) Energy: $0.02871 \times 931 = 26.7 [\text{MeV}]$ (1)		3		3	3	
	(c)		Benefit: routine supply of energy (i.e. does not depend on weather) or jobs or no CO₂ emission (1) Issue: needs secure storage of radioactive waste (products) over a extended time period or reference to long half-lives or long build time or building extra transmission power lines (1) Reasoned conclusion (1)			3	3		
			Question 2 total	4	3	3	10	3	0